# DATABASE MANAGEMENT SYSTEM LAB FILE

NAME : SUMIT KUMAR SINHA

SECTION : 2CSE5

ROLL NO : 2410030721

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## EXPERIMENT : 01

1. Table as per given constraints

TABLE : Employee

| COLUMN NAME | DATA TYPE | SIZE | ATTRIBUTES |
|---|---|---|---|
| EMPNO | NUMBER | 4 | PRIMARY KEY |
| ENAME | VARCHAR2 | 20 | NOT NULL |
| JOB | VARCHAR2 | 20 | |
| MGR | NUMBER | 4 | |
| HIREDATE | DATE | | |
| SAL | NUMBER | 10 | |
| COMM | NUMBER | 7 | |
| DEPTNO | NUMBER | 2 | FOREIGN KEY |

TABLE : DEPARTMENT

| COLUMN NAME | DATA TYPE | SIZE |
|---|---|---|

| | | |
|---|---|---|
| DEPTNO | NUMBER | 2 |
| DNAME | VARCHAR2 | 15 |

VALUES IN THE TABLE

| EMPNO | ENAME | JOB | MGR | HIREDATE | SAL | COMM | DEPTNO |
|---|---|---|---|---|---|---|---|
| 7369 | SMITH | CLERK | 7902 | 17-DEC-80 | 800 | | 20 |
| 7499 | ALLEN | SALESMAN | 7698 | 20-DEC-81 | 1600 | 300 | 30 |
| 7521 | WARD | SALESMAN | 7698 | 22-FEB-81 | 1250 | 300 | 30 |
| 7566 | JONES | MANAGER | 7839 | 02-APRIL-81 | 2975 | | 20 |
| 7654 | MARTI | SALESMAN | 7698 | 28-SEP-81 | 1250 | 1400 | 30 |
| 7698 | BLAKE | MANAGER | 7839 | 01-MAY-81 | 2850 | | 30 |
| 7782 | CLARK | MANAGER | 7839 | 09-JUN-81 | 2450 | | 20 |
| 7788 | SCOTT | ANALYST | 7566 | 09-DEC-82 | 3000 | | 40 |
| 7839 | KING | PRESIDENT | | 17-NOV-81 | 5000 | | 20 |
| 7844 | TURNER | SALESMAN | 7698 | 08-SEP-81 | 1500 | 0 | 30 |
| 7876 | ADAMS | CLERK | 7788 | 12-JAN-83 | 1100 | | 20 |
| 7900 | JAMES | CLERK | 7698 | 03-DEC-81 | 950 | | 30 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 7902 | FORD | ANALYST | 7566 | 03-DEC-81 | 3000 | | 20 |
| 7934 | MILLER | CLERK | 7782 | 23-JAN-82 | 1300 | | 10 |

| DEPTNO | DNAME |
|---|---|
| 10 | RESEARCH |
| 20 | ACCOUNTING |
| 30 | SALES |
| 40 | OPERATIONS |

PERFORM FOLLOWING QUERY

1. CREATE EMPLOYEE_MASTER TABLE WITH DATA USING EMPLOYEE

```
MariaDB [sumit]> CREATE TABLE EMPLOYEE_MASTER AS
    -> SELECT * FROM EMPLOYEE;
Query OK, 14 rows affected (0.045 sec)
Records: 14  Duplicates: 0  Warnings: 0

MariaDB [sumit]> SHOW TABLES;
+-----------------+
| Tables_in_sumit |
+-----------------+
| department      |
| employee        |
| employee_master |
+-----------------+
3 rows in set (0.001 sec)

MariaDB [sumit]> USE EMPLOYEE MASTER;
ERROR 1049 (42000): Unknown database 'employee'
MariaDB [sumit]> SELECT * FROM EMPLOYEE_MASTER;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  880 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 3273 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 3135 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2695 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3300 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5500 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1210 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 | 1045 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3300 | NULL |     20 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1430 | NULL |     10 |
+-------+--------+-----------+------+------------+------+------+--------+
14 rows in set (0.001 sec)
```

## 2) DELETE ALL RECORD INTO EMPLOYEE_MASTER WHOSE DEPTNO IS 10

```
MariaDB [sumit]> DELETE FROM EMPLOYEE_MASTER
    -> WHERE DEPTNO = 10;
Query OK, 1 row affected (0.010 sec)

MariaDB [sumit]> SELECT * FROM EMPLOYEE_MASTER;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  880 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 3273 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 3135 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2695 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3300 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5500 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1210 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 | 1045 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3300 | NULL |     20 |
+-------+--------+-----------+------+------------+------+------+--------+
13 rows in set (0.006 sec)

MariaDB [sumit]>
```

3) Update 10% in his salary of DEPTNO 20 into EMPLOEYEE_MASTER

```
MariaDB [sumit]> UPDATE EMPLOYEE_MASTER
    -> SET SAL = SAL + (SAL * 0.10)
    -> WHERE DEPTNO = 20;
Query OK, 6 rows affected (0.015 sec)
Rows matched: 6  Changed: 6  Warnings: 0

MariaDB [sumit]> SELECT * FROM EMPLOYEE_MASTER;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  968 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 3600 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 3135 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2965 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3300 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 6050 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1331 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 | 1045 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3630 | NULL |     20 |
+-------+--------+-----------+------+------------+------+------+--------+
13 rows in set (0.001 sec)

MariaDB [sumit]> |
```

4) Alter SAL with size 10,2 in EMPLOYEE_MASTER.

```
MariaDB [sumit]> ALTER TABLE EMPLOYEE_MASTER
    -> MODIFY SAL DECIMAL(10,2);
Query OK, 13 rows affected (0.086 sec)
Records: 13  Duplicates: 0  Warnings: 0

MariaDB [sumit]> SELECT * FROM EMPLOYEE_MASTER;
+-------+--------+-----------+------+------------+---------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL     | COMM | DEPTNO |
+-------+--------+-----------+------+------------+---------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  968.00 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600.00 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250.00 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 3600.00 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250.00 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 3135.00 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2965.00 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3300.00 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 6050.00 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500.00 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1331.00 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 | 1045.00 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3630.00 | NULL |     20 |
+-------+--------+-----------+------+------------+---------+------+--------+
13 rows in set (0.002 sec)

MariaDB [sumit]> |
```

## 5) DROP EMPLOYEE_MASTERTABLE.

```
MariaDB [sumit]> DROP TABLE EMPLOYEE_MASTER;
Query OK, 0 rows affected (0.027 sec)

MariaDB [sumit]> SHOW TABLES;
+-----------------+
| Tables_in_sumit |
+-----------------+
| department      |
| employee        |
+-----------------+
2 rows in set (0.001 sec)

MariaDB [sumit]> |
```